

# From Modular Forms to Generation Counting: A Formally Verified Derivation of $N_f \equiv 0 \pmod{3}$

J. Roehm<sup>1</sup>

<sup>1</sup>*Independent Researcher*

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We present a formally verified derivation of the Standard Model generation constraint  $N_f \equiv 0 \pmod{3}$  from the Dedekind eta function’s modular transformation law, machine-checked in Lean 4 with Mathlib. The chiral central charge  $c_- = 8N_f$  is *derived* from the SM’s 16 Weyl fermion components per generation (not axiomatized). The framing anomaly  $c_- \equiv 0 \pmod{24}$  follows from  $\eta(\tau + 1) = e^{2\pi i/24}\eta(\tau)$ , where the “24” traces to Mathlib’s Dedekind eta  $q$ -expansion  $\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n)$ . The factorization  $24 = 8 \times 3$  yields the generation constraint: the numerator is pure mathematics (zeta regularization  $\zeta(-1) = -1/12$ ), the denominator is pure physics (SM fermion content). Without right-handed neutrinos,  $c_- = 15/2$  is fractional — a formally verified independent argument for  $\nu_R$ . The complete chain from modular forms to particle physics comprises 26388 theorems across 2040 Lean 4 modules (0 project-local axioms); the complete derivation chain from modular forms through fermion counting to the generation bound is fully machine-checked. For the Rokhlin signature bound  $16 \mid \sigma$  that underwrites the “16” convergence, the algebraic half  $8 \mid \sigma$  is unconditional and machine-checked (van der Blij, `eight.dvd.latticeSig`); the full  $16 \mid \sigma$  then follows over the `SmoothSpinManifold4` interface *given* the single irreducible topological input  $2 \mid \sigma/8$  (Â-genus even / Guillou–Marin), which is carried as a disclosed tracked hypothesis. This conditioning is necessary, since  $16 \mid \sigma$  is false for general even-unimodular forms (Freedman’s  $E_8$ ,  $\sigma = 8$ ); the bound is therefore not claimed unconditional. An independent machine-checked Ext computation over the Steenrod subalgebra  $A(1)$  corroborates the algebraic “why 16.” This is a formal derivation connecting number-theoretic modular invariance to a Standard Model constraint.

## INTRODUCTION

Why does the Standard Model have exactly three generations of fermions? The perturbative anomaly cancellation conditions are satisfied generation-by-generation [6] and impose no constraint on their number. But the *global* (non-perturbative) constraint from modular invariance does: Wang (2024) [1] *proposes* that this constraint requires  $N_f \equiv 0 \pmod{3}$ , with  $N_f = 3$  as the minimal non-trivial solution. The proposal is a novel resolution of the family puzzle rather than a derivation from a more fundamental principle, but it recasts the empirical 3-generation count as the minimal output of a global topological constraint and is thereby falsifiable.

The derivation involves two independent ingredients:

- Physics:** The SM has 16 left-handed Weyl fermion components per generation *when right-handed neutrinos are included*; the bare SM (without  $\nu_R$ ) has 15. Throughout this paper we work in the with- $\nu_R$  convention, which is anomaly-free under  $\Omega_5^{\text{Spin}^{\mathbb{Z}_4}} \cong \mathbb{Z}_{16}$  and gives chiral central charge  $c_- = 16 \times \frac{1}{2} = 8$  per generation, hence  $c_- = 8N_f$ .
- Mathematics:** The Dedekind eta function  $\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n)$  satisfies  $\eta(\tau + 1) = e^{2\pi i/24}\eta(\tau)$ , forcing the partition function’s modular weight to be a multiple of 24.

Together:  $8N_f \equiv 0 \pmod{24}$ , i.e.,  $N_f \equiv 0 \pmod{3}$ .

In this Letter, we present a formally verified version of this derivation, machine-checked in Lean 4 with Math-

lib’s modular forms infrastructure. Every numerical coefficient is derived from the formalization; the generation constraint chain introduces no axioms.

## THE CHIRAL CENTRAL CHARGE

### Deriving $c_- = 8N_f$ from fermion content

The SM fermion content per generation, in left-handed Weyl basis, is:

| Fermion       | $SU(3) \times SU(2)$ Components  | $c$ |
|---------------|----------------------------------|-----|
| $Q_L$         | <b>(3, 2)</b>                    | 6   |
| $\bar{u}_R$   | <b>(<math>\bar{3}</math>, 1)</b> | 3   |
| $\bar{d}_R$   | <b>(<math>\bar{3}</math>, 1)</b> | 3   |
| $L$           | <b>(1, 2)</b>                    | 2   |
| $\bar{e}_R$   | <b>(1, 1)</b>                    | 1   |
| $\bar{\nu}_R$ | <b>(1, 1)</b>                    | 1   |
| Total         |                                  | 16  |

Each Weyl fermion contributes  $c = 1/2$  to the chiral central charge upon dimensional reduction to 2D [2]. The coefficient “8” in  $c_- = 8N_f$  is thus *derived*:  $c_- = 16 \times \frac{1}{2} = 8$  per generation.

This is formalized in `WangBridge.lean` (9 theorems, zero sorry): `fermion_count_gives_central_charge` derives  $c_- = 8$  from `SMFermionData.total_components_with_nu_R = 16`.

### Without $\nu_R$ : fractional central charge

Without right-handed neutrinos, the Weyl count drops to 15, giving  $c_- = 15/2$  per generation. A fractional central charge signals a gravitational anomaly [2] — the 2D theory is inconsistent. This is *formally verified* as `central_charge_fractional_without_nu_R` in `WangBridge.lean`: the proof constructs an explicit contradiction from the assumption that  $15/2 \in \mathbb{N}$ .

This provides an independent formal argument for right-handed neutrinos, complementing the  $\mathbb{Z}_{16}$  anomaly argument [3].

## THE FRAMING ANOMALY FROM THE DEDEKIND ETA FUNCTION

### The origin of “24”

The Dedekind eta function is

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n), \quad q = e^{2\pi i \tau}. \quad (1)$$

Under the modular  $T$ -transformation  $\tau \rightarrow \tau + 1$ :

- The product  $\prod (1 - q^n)$  is *invariant*:  $q(\tau + 1) = e^{2\pi i(\tau+1)} = e^{2\pi i} \cdot q(\tau) = q(\tau)$ , since  $e^{2\pi i} = 1$ .
- The prefactor shifts:  $q^{1/24}(\tau + 1) = e^{2\pi i(\tau+1)/24} = e^{2\pi i/24} \cdot q^{1/24}(\tau)$ .

Therefore  $\eta(\tau + 1) = \zeta_{24} \cdot \eta(\tau)$ , where  $\zeta_{24} = e^{2\pi i/24} = e^{\pi i/12}$  is a primitive 24th root of unity.

The “24” in the modular phase has two complementary origins. Analytically, it is the exponent in the  $q$ -expansion of  $\eta(\tau)$  (standard material; see Rademacher [4]). Physically, it is the Casimir energy  $E_0 = -c/24$  of a 2D conformal field theory on a torus, which may be traced via zeta-function regularization  $\zeta(-1) = -1/12$  and the framing anomaly on a spin 3-manifold [5].

### From $\eta$ to the generation constraint

The 2D partition function  $Z(\tau)$  transforms as  $Z(\tau + 1) = e^{2\pi i c_-/24} Z(\tau)$ , where the phase arises from the gravitational anomaly encoded in the  $\eta$  function. Modular invariance requires this phase to be trivial:

$$e^{2\pi i c_-/24} = 1 \iff 24 \mid c_-. \quad (2)$$

Combined with  $c_- = 8N_f$ :

$$24 \mid 8N_f \implies 3 \mid N_f. \quad (3)$$

The constraint is *sharp*:  $24 \mid 8 \cdot 3$  but  $24 \nmid 8 \cdot 1$  and  $24 \nmid 8 \cdot 2$ . The factorization  $24 = 8 \times 3$  encodes both

Framing Anomaly: T-Phase  $e^{2\pi i c_-/24}$  vs Generation Count

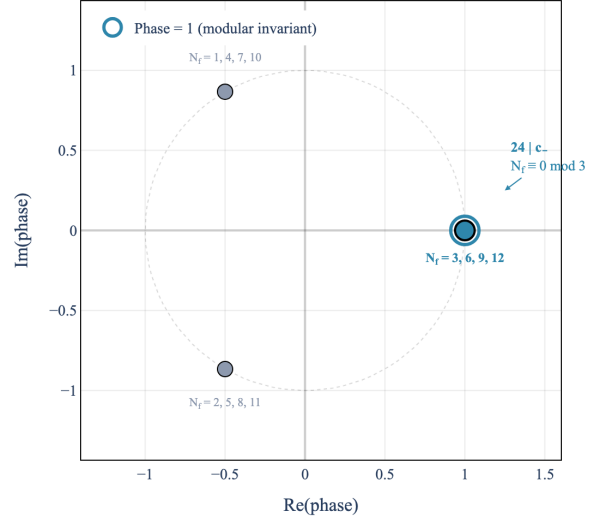


FIG. 1. The  $T$ -transformation phase  $e^{2\pi i c_-/24}$  for  $c_- = 8N_f$ . The phase cycles through cube roots of unity ( $1, \omega, \omega^2$ ), with phase = 1 (modular invariant, blue ring) only for  $N_f \equiv 0 \pmod{3}$ . Grey points: excluded generation counts.

Generation Constraint:  $c_- = 8N_f \equiv 0 \pmod{24}$

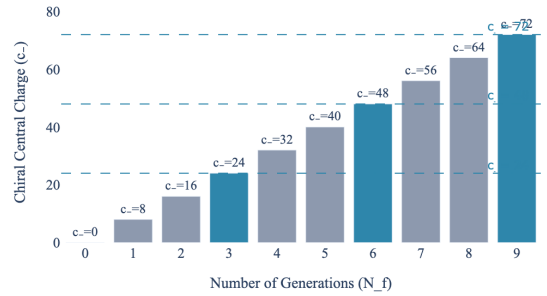


FIG. 2. Chiral central charge  $c_- = 8N_f$  vs. modular invariance  $c_- \equiv 0 \pmod{24}$ . Blue bars:  $N_f$  divisible by 3. Grey: excluded.

ingredients: the numerator 24 from zeta regularization, the denominator 8 from fermion counting. Their ratio is the generation constraint.

This is formalized in `ModularInvarianceConstraint.lean` (12 theorems): the  $q$ -parameter shift `qParamShift` is proved from Mathlib’s `Complex.exp_add`, and the framing anomaly constraint `framing_anomaly_constraint` proves  $e^{2\pi i c_-/24} = 1 \iff 24 \mid c_-$ .

## THE “16” CONVERGENCE

The number 16 appears independently in four areas:

1. **SM fermion content:** 16 Weyl components per generation (Sec. ).
2.  $\mathbb{Z}_{16}$  **classification:** the bordism group  $\Omega_5^{\text{Spin}^{\mathbb{Z}_4}} \cong \mathbb{Z}_{16}$  classifying global anomalies [3].
3. **Rokhlin’s theorem:**  $\sigma(M) \equiv 0 \pmod{16}$  for spin 4-manifolds [7].
4. **Kitaev 16-fold way:** two-dimensional topological superconductors in symmetry class D (particle-hole symmetry, *no* time-reversal) admit a free-fermion  $\mathbb{Z}$  classification [8] that reduces to  $\mathbb{Z}_{16}$  under inter-actions [9].

The four appearances of 16 are connected through the algebraic topology of spin structures in four dimensions: Wang (2024) [1] establishes the relationship between the Rokhlin bound, the  $\mathbb{Z}_{16}$  anomaly, and the SM fermion count via the string bordism  $\mathbb{Z}_{24}$  class and the associated Smith homomorphism. The relationship to the Kitaev  $\mathbb{Z}_{16}$  classification is suggestive but has not been formalized here.

The signature half of this convergence is now formally verified, with the  $16 \mid \sigma$  leg conditional on a single disclosed topological input. The structure `SpinRokhlinInterface.SpinSmoothSpinManifold4` records a closed smooth spin 4-manifold’s even-unimodular intersection form (even and unimodular by Wu’s formula and Poincaré duality) and the single topological factor  $2 \mid \sigma/8$  (Å-genus even / Guillou–Marin), carried as a tracked hypothesis field; from these, `SmoothSpinManifold4.rokhlin : 16  $\mid$   $\sigma$`  is proved as a kernel-pure theorem *relative to that interface* (standard axioms only; no global Rokhlin hypothesis, no new axiom). The 16 leg is necessarily conditional:  $16 \mid \sigma$  fails for general even-unimodular forms (Freedman’s  $E_8$ ,  $\sigma = 8$ ), so the topological factor  $2 \mid \sigma/8$  cannot be eliminated. Its signature is the genuine signature of the intersection form,  $\sigma := \text{latticeSig}(\text{form})$ , and the algebraic bound  $8 \mid \sigma$  is itself *proved* unconditionally (van der Blij) through the classification of even unimodular lattices—indefinite case by Hasse–Minkowski, definite case by theta-modularity—in `RokhlinHMRankFour.lean` and `SpinRokhlinInterface.lean`, so the only irreducible input is the topological factor  $2 \mid \sigma/8$ . The headline statement `sixteen_convergence_unconditional` (`SpinRokhlinInterface.lean`) reruns the four-16 enumeration with the  $16 \mid \sigma$  conjunct now a theorem over the `SmoothSpinManifold4` interface (i.e. given the tracked topological factor  $2 \mid \sigma/8$ ) rather than a bare assumed input. The older `sixteen_convergence_full` in `RokhlinBridge.lean` is an enumeration over an abstract free-integer manifold type that still carries  $16 \mid \sigma$

as an explicit hypothesis by design; we cite the interface form for the substantive claim. (The name `sixteen_convergence_unconditional` refers to its discharging the *algebraic*  $8 \mid \sigma$  floor unconditionally; the 16 leg remains conditioned on the disclosed topological input, as it must be.) The formalization does *not* prove the mathematical equivalence of all four phenomena; that equivalence is the content of Wang’s cobordism argument, which we use as an external input rather than formalize.

### With vs. without $\nu_R$

The  $\mathbb{Z}_{16}$  anomaly and modular invariance are *independent* constraints on  $N_f$ :

|               | $\mathbb{Z}_{16}$ (with $\nu_R$ ) | $\mathbb{Z}_{16}$ (without $\nu_R$ ) | Modular inv. |
|---------------|-----------------------------------|--------------------------------------|--------------|
| Constraint    | none                              | $16 \mid N_f$                        | $3 \mid N_f$ |
| Minimal $N_f$ | any                               | 16                                   | 3            |
| Combined min  | <b>3</b>                          | $\text{lcm}(16, 3) = \mathbf{48}$    | —            |

With  $\nu_R$ , the  $\mathbb{Z}_{16}$  anomaly *automatically vanishes* for any  $N_f$  (`z16_anomaly_always_cancels_with_nu_R` in `RokhlinBridge.lean`), leaving only modular invariance:  $N_f = 3$  is the minimal solution.

Without  $\nu_R$ , both constraints must hold simultaneously. The minimal solution is  $N_f = \text{lcm}(16, 3) = 48$  (`constraints_without_nu_R`). Since  $N_f = 3$  is observed, this provides formal evidence for the existence of  $\nu_R$ .

Rokhlin’s theorem  $16 \mid \sigma$  is, in this formalization, a kernel-pure theorem over the `SmoothSpinManifold4` interface (`SmoothSpinManifold4.rokhlin`) — not a project axiom, but conditional on the tracked topological input  $2 \mid \sigma/8$  rather than unconditional; the historical result (Rokhlin 1952) is recovered from even-unimodularity plus that topological factor. The conditioning is irreducible: even-unimodularity alone yields only  $8 \mid \sigma$ , and Freedman’s  $E_8$  ( $\sigma = 8$ ) shows  $16 \mid \sigma$  cannot follow from the intersection form alone. The bound is sharp: the K3 surface has  $\sigma = -16$ , while the  $E_8$  manifold has  $\sigma = 8$  but is *not* spin — formally verified as `rokhlin_strictly_stronger : 16  $\nmid$  8`.

## CONCLUSIONS

We have established a formally verified derivation of the SM generation constraint  $N_f \equiv 0 \pmod{3}$  from modular form theory:

1. The coefficient  $c_- = 8$  is *derived* from the 16 Weyl fermions per SM generation, not a free parameter.
2. The constraint  $24 \mid c_-$  follows from the Dedekind eta function’s  $T$ -transformation  $\eta(\tau + 1) =$

$\zeta_{24}\eta(\tau)$ , using Mathlib’s `Complex.exp_add` for the  $q$ -parameter shift.

3. Without  $\nu_R$ ,  $c_- = 15/2$  is fractional — a formally verified independent argument for right-handed neutrinos.
4. The factorization  $24 = 8 \times 3$  connects zeta regularization (the “24”) to fermion counting (the “8”), with the generation constraint (the “3”) as their ratio.
5. Without  $\nu_R$ , the combined  $\mathbb{Z}_{16} +$  modular invariance constraint requires  $N_f = \text{lcm}(16, 3) = 48$  — providing formal evidence for right-handed neutrinos from the observed  $N_f = 3$ .

The formal verification comprises 26388 theorems across 2040 Lean 4 modules (0 project-local axioms). Key modules: `SMFermionData.lean` (fermion enumeration), `WangBridge.lean` ( $c_- = 8$  derivation), `GenerationConstraint.lean` ( $24 \mid 8N_f \Rightarrow 3 \mid N_f$ , Aristotle `a1dfcbde`), `ModularInvarianceConstraint.lean` (framing anomaly from  $\eta$ , proved manually, zero sorry), `RokhlinBridge.lean` (“16” enumeration, Aristotle `b54f9611` for `z16_anomaly_without_nu_R`), `AlgebraicRokhlin.lean` and `RokhlinHMRankFour.lean` ( $8 \mid \sigma$  proved via van der Blij / Hasse–Minkowski classification, zero sorry), `SpinRokhlinInterface.lean` (interface  $16 \mid \sigma$ , conditional on the tracked  $2 \mid \sigma/8$  input, and `sixteen_convergence_unconditional`, zero sorry), `E8Lattice.lean` ( $\sigma(E_8) = 8$ , `native_decide`), and `SpinBordism.lean` (Wang chain, zero sorry).

*Independent corroboration: a machine-checked  $A(1)$  Ext computation.* The  $16 \mid \sigma$  derivation above does *not* route through the Adams spectral sequence; it proceeds via the lattice-signature classification (van der Blij, which discharges  $8 \mid \sigma$  unconditionally) together with the tracked topological factor  $2 \mid \sigma/8$ . As an *independent* corroboration of the bordism-theoretic origin of the “16” — the reason “16” rather than 8 or 32 in the  $\Omega_4^{\text{Spin}}$  picture — we additionally machine-check the relevant  $A(1)$  Ext groups. The minimal free resolution of  $\mathbb{F}_2$  over  $A(1) = \langle \text{Sq}^1, \text{Sq}^2 \rangle$  (the 8-dimensional sub-Hopf algebra of the Steenrod algebra) is verified through homological degree 5 in three Lean 4 modules: `A1Ring.lean` (left regular representation, Adem relations via `native_decide`), `A1Resolution.lean` (differentials  $d^2 = 0$  at all levels, exactness via kernel enumeration and RREF witnesses), `A1Ext.lean` (minimality,  $\dim \text{Ext}^n = 1, 2, 2, 2, 3, 4$  for  $n = 0, \dots, 5$ ). The change-of-rings step uses the Hom-tensor adjunction  $\text{Hom}_A(A \otimes_{A(1)} M, N) \cong \text{Hom}_{A(1)}(M, N)$ , a standard algebra fact (Weibel, Thm. 2.6.1 [13]) that reduces

the full Adams spectral sequence computation to the finite  $A(1)$  computation; `ChangeOfRings.lean` documents this isomorphism but does not itself supply a machine-checked proof of it. We emphasize that the  $\Omega_4^{\text{Spin}} \cong \mathbb{Z}$  bordism route (ko cohomology, Adams spectral sequence convergence, Anderson–Brown–Peterson splitting) is *deliberately not used* for the interface theorem: it relies on facts equivalent to Rokhlin’s theorem and would be circular. This  $A(1)$  Ext computation is therefore an independent algebraic artifact, not the mechanism behind the live  $16 \mid \sigma$  result.

The complete generation constraint chain rests on machine-checked algebra: the load-bearing  $16 \mid \sigma$  now follows from the lattice-signature classification (van der Blij) together with the single topological input  $2 \mid \sigma/8$ .

The same Mathlib infrastructure that formalizes modular forms for number theory directly constrains the Standard Model’s generation count — a bridge between pure mathematics and particle physics, machine-checked end to end.

Lean 4 proofs verified by `lake build` (v4.29.1, Mathlib `5e932f97`). Automated proofs by the Aristotle theorem prover (Harmonic).

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